

Industrial Automation Software

To simplify the workflow and increase the efficiency and productivity

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In industrial automation, control systems integrating hardware and software are used for various machineries, reducing the need for human intervention. The control system comprises supervisory control and data acquisition (SCADA), distributed control systems (DCS), programmable logic controllers (PLCs), human-machine interfaces (HMIs), etc. Each control system or combination of them finds applications in various places like manufacturing, power generation and water treatment.

Wong Foot Yow, business development manager, Exava Sdn Bhd, says, "A basic SCADA system comprises mimic display for status monitoring and control, alarm for alerting, trending for data logging, report generation for management, etc. The most essential part of a SCADA system is the capability to communicate or interact with external devices, also known as the driver modules."

The advantages of industrial automation software are many and SCADA has almost become a mandatory system for today's factories. The only disadvantage could be the initial cost involved and the maintenance needed.

Given below are some of the industrial automation software that are included in this month's EFY-CD.

IntegraXor

IntegraXor is a robust SCADA server and a friendly, flexible and versatile project editor. The latest version available is 3.2.3786.4.

What's special? IntegraXor offers lots of features to automate and simplify the workflow and also improve efficiency and productivity. It offers full-blown cross-browser animation engine over the Internet with W3C-compliant technologies. The imple-

mentation concept is rather simple and easy-to-understand for newcomers. The innovative patent-pending report module offers a one-time configuration on layout only, instead of configuring database, tag entry and then working on layout. The multiple redundancy module maximises the processing power and network bandwidth.

Distributor. Payment accepted via credit card or T.T. for worldwide users.
Home Page: www.integraxor.com
Download URL: www.integraxor.com/download.htm
Platform Supported: Windows

MODbus PLC Simulator

MODbus PLC simulator can be used for process control and monitoring devices. The latest version of the software is v8.18, distributed under GPL licence 3.

What's special? Typically, the simulator can be used to verify a SCADA/HMI configuration, develop modbus master and slave drivers for embedded devices and learn the modbus protocols. It can simulate noisy, long, serial transmission lines. The noise simulator gobbles and injects bad characters, which helps developers test under harsh electrical noise environments without having to sit next to an electric arc furnace.

Distributor. Being free and open source, the software can be downloaded from the website.

Home Page: www.plcsimulator.org/
Download URL: www.plcsimulator.org/downloads
Platform Supported: Windows

Classic Ladder

Classic Ladder is a free PLC simulator. The latest version of the software available is 0.8.008, released under LGPL licence.

What's special? Classic ladder is

coded in 'C' and basically meant for educational purpose. It is split into two sections: ladder and sequential. Within ladder, the software offers Boolean elements, rising/falling edges, timers, monostables, etc. It also runs in real time with RTLinux v3, RTAI and now Xenomai. If you do not have any clue of how PLC works, visit www.plcs.net/contents.shtml

Distributor. Being free and open source, the software is available for download from sourceforge.net
Home Page: <http://membres.multimania.fr/mavati/classicladder/>
Download URL: <http://sourceforge.net/projects/classicladder/>
Platform Supported: Windows and Linux

Free SCADA

Free SCADA is an open source software available for Windows. The software is distributed under Common Public Licence 1.0.

What's special? It provides end-users flexible tools for visualisation and interactive control of any industrial process. It is written in C# and C++. The system uses OPC servers for data collection. The software has come a long way since its launch and now features an import graphics command, perfect runtime communication debugging, more scripting features, support for more examples for scripting feature and support for dynamically loaded visual plugins.

The minimum system requirements are Windows NT through XP and .NET Framework 3.5.

Distributor. Being free and open source, the software is available for download from sourceforge.net

Home Page: <http://free-scada.org/>
Download URL: <http://sourceforge.net/projects/free-scada/files/>
Platform Supported: Windows ●

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